Data Analytics Service Comparison - Michaël Bettan

BigQuery: Serverless (no-ops), highly-scalable and cost-effective multi-cloud Data Warehouse where you can run SQL queries over vast amounts of data	**Ideal:** • OLAP workloads • Data Warehousing • Structured data • Immediate consistency • Optimized for columnar storage and large-scale reads	**Avoid:** • Low latency reqs. • Non-structured data • Non-relational SQL • Store large objects
Dataflow: Apache Beam Managed Service for unified stream and batch data processing patterns. Automated infrastructure provisioning and cluster management with horizontal autoscaling.	**Ideal:** • Streaming Analytics • Real-time AI for Predictive Analytics or Fraud Detection • Sensor and log data process for System Health • ETL pipelines • Unified stream and batch processing	**Avoid:** • Wrangling capability • Low Data Volume or Velocity • Spark-exclusive workloads • Legacy tooling/teams (against change) • Very low latency (< seconds) • Small datasets • Simpler Data Processing needs
Dataproc: Managed Apache Spark and Hadoop service. Fast and easy-to-use. Dataproc provisions big and small clusters rapidly, supports many popular job types, and is integrated with other Google Cloud services (e.g., GCS, Stackdriver)	**Ideal:** • OSS Hadoop ecosystem tools (Pig, Hive, Spark, etc.). • Large-scale batch processing • Complex data transformations • Machine Learning with Spark	**Avoid:** • Wrangling capability • Serverless alternatives are preferred (cluster mgmt) • Workloads where managed services are preferred due to simplicity (e.g., BigQuery for SQL) • Lack of Python/Java expertise in the team • Small datasets • Simpler Data Processing needs
Pub/Sub: Event-driven asynchronous messaging service that decouples senders (producing events) and receivers (processing events). Allows for secure and highly available communication between independently written applications. Unified messaging, global presence, push & pull-style subscriptions, replicated storage and guaranteed at-least-once message delivery, encryption of data at rest/transit, easy-to-use REST API	**Ideal:** • Data streaming from various processes or devices • Balancing workloads in network clusters • Queue can efficiently distribute tasks • Implementing asynchronous workflows • Reliability improvement- in case zone failure • Distributing event notifications • Refreshing distributed caches • Logging to multiple systems	**Avoid:** • Kafka requirements • Budget constraint • Strict Ordering Requirements Across Multiple Publishers • Very large messages (prefer Cloud Storage for payload) • Synchronous request-response scenarios • Exactly-once delivery guarantees • Vendor-lock-in concerns • Rich Ecosystem and Integrations • High Throughput and Low Latency
Cloud Storage: Global, secure and scalable object store. Fully-managed and highly-reliable object/blob storage.	**Ideal:** • Eventual consistency • Objects and blobs • Storing unstructured data (images, videos, text, logs, backups) • Data lake implementation • Cost-effective storage for large dataset • Serving static website content	**Avoid:** • IoT • Spiky streaming data

Cloud SQL: Fully-managed relational database service supporting MySQL, PostgreSQL and MS SQL Server. Designed for relational data (record-based): tables, rows and columns, and super structured data. SQL compatible and can update fields.	**Ideal:** • SQL DB scaling vertically for structured data • Relational data (tables, rows, columns) • OLTP workloads • Structured data • MySQL, PostgreSQL, SQL Server compatibility • Managed database service	**Avoid:** • RDBMS + large scale OLAP workloads • NoSQL data • Extremely high throughput • Low-latency transactions • Databases larger than 30TB • non-relational, highly nested, or unstructured (e.g., documents, graphs, key-value pairs). • Massive Global Scalability Needs • Strict Horizontal Scaling • Fully serverless database
Cloud Spanner: Real time transaction store, horizontally autoscale and always available.	**Ideal:** • Relational database • Structured data • Vertical + Horizontal scaling • Strong consistency • Transactional reads and writes → mission critical workloads	**Avoid:** • Budget constraint • Data that is not relational or structured • Open-source RDBMS preference • When strong consistency and high availability are not critical • Small-Scale or Low-Traffic Applications • Vendor Lock-In Concerns • Lack of Need for Strong Consistency and ACID Transactions
AlloyDB for Postgres: A fully managed PostgreSQL-compatible database service for your most demanding enterprise workloads. AlloyDB combines the best of Google with PostgreSQL, for superior performance, scale, and availability.	**Ideal:** • PostgreSQL compatible • High performance OLTP • Automated scaling • Reduced operational overhead	**Avoid:** • Non-relational data • Workloads not requiring high throughput
Dataplex: Intelligent data fabric that helps organize, manage, and govern data across Google Cloud services.	**Ideal:** • Data discovery and metadata management • Data lineage tracking • Data quality monitoring • Data governance and access control	**Avoid:** • Projects with limited data complexity • Lack of need for centralized data management
Bigtable: A managed, scalable NoSQL, wide column database service	**Ideal:** • PB Large analytical workloads (OLAP) • OLAP and OLTP workloads • Real time data • High throughput analysis & Low latency writes • Key-based reads • Time-series workloads (Financial, IoT, etc.) • HBase legacy migration • Native time series support • NoSQL • PB Large OLAP workloads • Key-based reads • Schema free • Eventual consistency	**Avoid:** • OLTP workloads? • Relational data modeling • Replacement for MongoDB • Store large objects • Strong consistency requirements

	• Low-latency reads and writes of individual records within massive datasets. • Petabyte-scale storage with very fast key-value lookups. • high-throughput, low-latency	
Cloud Memorystore is a cache and better suited to storing key-value data for applications that need low latency access to data.	**Ideal:** • Caching • Session management • Real-time analytics • High-performance computing	**Avoid:** • Data persistence as primary requirement • Large datasets that don't fit in memory
Composer is a Managed Apache Airflow service for orchestrating workflows.	**Ideal:** • Data pipelines • ETL processes • Workflow automation	**Avoid:** • Simple, non-complex workflows • Real-time data processing
Data Fusion Fully managed, cloud-native data integration service.	**Ideal:** • ETL/ELT processes • Data pipelines • Data transformation	**Avoid:** • Simple data transformations • Limited budget
BigQueryML is Machine learning directly inside BigQuery.	**Ideal:** • Predictive analytics within BigQuery • Model training and deployment within BigQuery	**Avoid:** • Complex machine learning models requiring specialized hardware • Need for external ML tools
Cloud Firestore is a scalable, fully-managed, NoSQL document database.	**Ideal:** • Semi-structured data • Hierarchical data • NoSQL • Durable Key-value data • Schema free • OLTP workloads • Highly available apps requirements • ACID transactions	**Avoid:** • Not for Analytical workloads • Extreme scale • Sub-ms latency
Datastream is a serverless change data capture and replication service that lets you synchronize data stored in supported sources to a variety of destinations. This enables real-time analytics, database migration, and event-driven architectures with minimal operational overhead.	**Ideal:** • Highly available apps requirements • ACID transactions	**Avoid:** • Situations requiring complex transformations or joins before data lands in the destination
MongoDB: document-oriented database program. NoSQL database program, MongoDB uses JSON-like documents with optional schemas.	**Ideal:** • Flexible schema • Horizontal scalability • Document-oriented (suitable for various data types) • IoT data structures • low-latency reads	**Avoid:** • Transactions requiring strong ACID properties • Complex joins
Redis is an open-source, in-memory data store. It's often called a "data structures server" because it allows you to store and manipulate data in various formats like strings, lists, sets, and hashes.	**Ideal:** • In-memory data store caching • Real-time analytics • Session management • Leaderboards	**Avoid:** • Large datasets that don't fit in memory • Transactions requiring strong consistency (ACID properties)

	• Pub/Sub messaging	• Complex queries and data relationships • Persistent storage as the primary requirement
Cassandra: A highly scalable and distributed wide-column NoSQL database. It's designed to handle large volumes of data across multiple servers, providing high availability and fault tolerance. Cassandra prioritizes write performance and is well-suited for applications with massive data growth and unpredictable data structures.	**Ideal:** • High availability • Fault tolerance scalability • ACID transactions within a partition • time-series data (e.g., telemetry) • handle large volumes of data	**Avoid:** • Transactions spanning multiple partitions requiring strong consistency • Complex queries and joins across multiple partitions. • Strict ACID guarantees
Dataprep is intelligent cloud data service to visually explore, clean, and prepare data for analysis and machine learning	**Ideal:** • User-friendly Dataflow pipeline (no code) • Visual data exploration and cleaning • Data preparation for analysis and machine learning	**Avoid:** • Very large datasets that may exceed Dataprep limitations • Highly complex transformations requiring custom coding
Looker Studio : easy to use data visualization and dashboards	**Ideal:** • No budget (zero cost) • Easy-to-use data visualization and dashboards • Integration with various data sources • No budget (free to use)	**Avoid:** • Advanced analytical capabilities beyond visualization • Limited data transformation options within the tool
MariaDB: is a community-developed, commercially supported fork of the MySQL relational database management system, intended to remain free and open-source software under the GNU General Public License	**Ideal:** • Community-developed and commercially supported • Open-source (GNU GPL) • Relational database with SQL support • ACID compliant	**Avoid:** • Less scalable than some NoSQL solutions • Can be complex to manage for very large deployments
CouchDB: is a NoSQL, document-oriented database that stores data in JSON format, with a focus on scalability and availability. It uses Multi-Version Concurrency Control (MVCC) for consistency and offers a RESTful HTTP API to interact with the database.	**Ideal:** • Distributed environments: highly effective for distributed applications due to its built-in replication and synchronization across multiple nodes • Offline-first applications: CouchDB is ideal for applications that need to work offline, with automatic synchronization once connectivity is restored (e.g., mobile apps). • Document-based data (user profiles, product catalogs, or logs)	**Avoid:** • High-performance transactional systems: eventual consistency model is not suited for applications that require strong ACID-compliant transactions, like banking or inventory systems. • Complex querying and analytics: If you require complex queries, joins, or analytical workloads, relational databases or specialized analytical databases